Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/09/2007
PI Name:	Wade, Charles Ph.D.		
Project Title:	Alterations of Steroidogenesis by Exercise Countermeasures during Bed Rest		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78234-6315	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-05
Start Date:	04/30/2005	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008, per P. Denkins/JSC (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stein, Peter (University of Medicine and Dentistry of NJ - SOM) Tou, Janet (NASA Ames Research Center)		
Grant/Contract No.:	NNJ05HD37G		
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Task Description:	Countermeasures have been developed to alleviate some of the adverse effects of space flight. The end point for efficacy is usually the specific symptom of interest with little regard for confounding factors. Heavy exercise has been advocated as an effective countermeasure to attenuate loss of bone and muscle mass during space flight as well as to reduce the incidence of orthostatic intolerance upon return to earth. The secondary effects of heavy daily exercise have not been investigated. One of the primary effects of over training is a reduction in testosterone in males and a decrease in estrogen in females. These hormones have been implicated in the maintenance of normal bone and muscle health. We postulate that there are reductions in testosterone and estrogen during the use of heavy exercise as a countermeasure. In the proposed bed rest studies we hypothesize that the high levels of exercise required to sustain health of the musculoskeletal system will adversely effect steroidogenesis and thus, reproductive cycle. Further, in the previous study the level of exercise by males reduced testosterone levels. In both genders, a decrease in steroids if significant can contribute to loss of muscle and bone mass thereby negating the efficacy of exercise as a countermeasure. Hypothesis: The high levels of exercise necessary to attenuate muscle and bone loss in space flight (weightlessness) will adversely effect steroidogenesis leading to a changes in the female reproductive cycle. Specific Aims: 1) Estradiol and testosterone (urine and plasma) will be decreased in the female subjects due to a shift in steroidogenesis in the exercise countermeasure group compared to the control and drug treated groups. 2) Reproductive cycle, as determined by daily urine sample hormone levels and plasma concentrations at specific times in the cycle, will be altered (longer cycle) in the exercise countermeasure group compared to the control and drug treated groups. 3) Urinary testosterone excretion in male subjects will be decreased due to a shift in steroidogenesis in the exercise countermeasure group compared to the control and drug treated groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No report received this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2014)
Articles in Peer-reviewed Journals	Wade CE, Stanford KI, Stein TP, Greenleaf JE. "Intensive exercise training suppresses testosterone during bed rest." J Appl Physiol (1985). 2005 Jul;99(1):59-63. Epub 2005 Feb 10. PubMed PMID: 15705732 , Jul-2005